

Differentiation Rules Practice

Answer Key

AP Calculus AB/BC

Quick Answers

1. $20x^4 - 6x$

2. $6x^2 - \frac{5}{x^2}$

3. $9x^2 - 8x + 3$

4. $-\frac{7}{(x-3)^2}$

5. $\frac{1}{2\sqrt{x}} - \frac{18}{x^4}$

6. $15x^4 + 15x^2 - 2$

7. $\frac{2x}{(x^2+1)^2}$

8. (a) the derivative = $3x^2 + 4x - 5$, (b) the slope at $x = 2 = 15$ 9. (a) the derivative = $\frac{-2x^2+2x+6}{(x^2+3)^2}$, (b) the slope at $x = 1 = \frac{3}{8}$ 10. (a) the derivative = $\frac{x^2-2x-3}{(x-1)^2}$, (b) the slope at $x = 3 = 0$, —

What is verified

13 of 14 answers machine-checked against SymPy, across 10 problems.

— marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus AB/BC

FUN-3.B — problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–5 of 5 across 14 tagged checks.

Common wrong answers

8. If they got 4: multiplied the derivatives of the two factors instead of using the product rule

9. If they got 1: divided the derivative of the numerator by the derivative of the denominator

1. Differentiate $y = 4x^5 - 3x^2 + 7$.**Solution:** Power rule term by term, $\frac{d}{dx}x^n = nx^{n-1}$; the constant 7 has slope 0.

$$y' = 4(5)x^4 - 3(2)x^1 + 0 = 20x^4 - 6x.$$

$20x^4 - 6x$

2. Differentiate $y = 2x^3 + \frac{5}{x}$.

Solution: Rewrite $\frac{5}{x} = 5x^{-1}$ so the power rule applies:

$$y' = 6x^2 + 5(-1)x^{-2} = 6x^2 - \frac{5}{x^2}.$$

$$\boxed{6x^2 - \frac{5}{x^2}}$$

No quotient rule is needed — a constant over a power is just a power.

3. Differentiate $y = (x^2 + 1)(3x - 4)$.

Solution: Product rule, $(uv)' = u'v + uv'$ with $u = x^2 + 1$ and $v = 3x - 4$:

$$\begin{aligned} y' &= (2x)(3x - 4) + (x^2 + 1)(3) \\ &= 6x^2 - 8x + 3x^2 + 3 = 9x^2 - 8x + 3. \end{aligned}$$

So the derivative is

$$\boxed{9x^2 - 8x + 3}$$

4. Differentiate $y = \frac{2x + 1}{x - 3}$.

Solution: Quotient rule, $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$, with $u = 2x + 1$, $v = x - 3$:

$$y' = \frac{2(x - 3) - (2x + 1)(1)}{(x - 3)^2} = \frac{2x - 6 - 2x - 1}{(x - 3)^2} = \frac{-7}{(x - 3)^2}.$$

So the derivative is

$$\boxed{\frac{-7}{(x - 3)^2}}$$

The order $u'v - uv'$ matters: reversing it flips the sign to $+7$.

5. Differentiate $y = \sqrt{x} + \frac{6}{x^3}$.

Solution: Rewrite both terms as powers: $\sqrt{x} = x^{1/2}$ and $\frac{6}{x^3} = 6x^{-3}$.

$$y' = \frac{1}{2}x^{-1/2} + 6(-3)x^{-4} = \frac{1}{2\sqrt{x}} - \frac{18}{x^4}.$$

So the derivative is

$$\boxed{\frac{1}{2\sqrt{x}} - \frac{18}{x^4}}$$

6. Differentiate $y = (3x^2 - 1)(x^3 + 2x)$.

Solution: Product rule with $u = 3x^2 - 1$, $v = x^3 + 2x$:

$$\begin{aligned} y' &= (6x)(x^3 + 2x) + (3x^2 - 1)(3x^2 + 2) \\ &= 6x^4 + 12x^2 + (9x^4 + 6x^2 - 3x^2 - 2) \\ &= 15x^4 + 15x^2 - 2. \end{aligned}$$

So the derivative is

$$\boxed{15x^4 + 15x^2 - 2}$$

7. Differentiate $y = \frac{x^2}{x^2 + 1}$.

Solution: Quotient rule with $u = x^2$, $v = x^2 + 1$:

$$y' = \frac{(2x)(x^2 + 1) - x^2(2x)}{(x^2 + 1)^2} = \frac{2x^3 + 2x - 2x^3}{(x^2 + 1)^2} = \frac{2x}{(x^2 + 1)^2}.$$

So the derivative is

$$\frac{2x}{(x^2 + 1)^2}$$

8. $f(x) = (x + 2)(x^2 - 5)$. (a) $f'(x)$. (b) The slope at $x = 2$.

Solution (a): Product rule with $u = x + 2$, $v = x^2 - 5$:

$$f'(x) = (1)(x^2 - 5) + (x + 2)(2x) = x^2 - 5 + 2x^2 + 4x = 3x^2 + 4x - 5,$$

so

$$3x^2 + 4x - 5$$

Solution (b): Substitute $x = 2$: $3(4) + 4(2) - 5 = 12 + 8 - 5$.

$$15$$

Multiplying the two derivatives instead ($1 \cdot 2x = 2x$, giving 4 at $x = 2$) is the standard product-rule error; the number to watch for is 4.

9. $g(x) = \frac{2x - 1}{x^2 + 3}$. (a) $g'(x)$. (b) $g'(1)$.

Solution (a): Quotient rule with $u = 2x - 1$, $v = x^2 + 3$:

$$g'(x) = \frac{2(x^2 + 3) - (2x - 1)(2x)}{(x^2 + 3)^2} = \frac{2x^2 + 6 - 4x^2 + 2x}{(x^2 + 3)^2},$$

which collects to

$$\frac{-2x^2 + 2x + 6}{(x^2 + 3)^2}$$

Solution (b): At $x = 1$ the numerator is $-2 + 2 + 6 = 6$ and the denominator is $(1 + 3)^2 = 16$, so $g'(1) = \frac{6}{16}$, which reduces to

$$\frac{3}{8}$$

Dividing derivative-by-derivative instead would give $\frac{2}{2x} = 1$ at $x = 1$; the number to watch for is 1.

10. Challenge. $f(x) = \frac{x^2 + 3x}{x - 1}$. (a) $f'(x)$. (b) $f'(3)$. (c) What does (b) say about the graph?

Solution (a): Quotient rule with $u = x^2 + 3x$, $v = x - 1$:

$$f'(x) = \frac{(2x + 3)(x - 1) - (x^2 + 3x)(1)}{(x - 1)^2} = \frac{2x^2 + x - 3 - x^2 - 3x}{(x - 1)^2},$$

so the simplified derivative is

$$\frac{x^2 - 2x - 3}{(x - 1)^2}$$

Solution (b): At $x = 3$ the numerator is $9 - 6 - 3 = 0$ and the denominator is $(3 - 1)^2 = 4$, so the slope is 0

Solution (c) — instructor-judged, no machine check. A zero derivative means the tangent line at that point is **horizontal**, so the graph of f levels off at $x = 3$. Full credit says horizontal tangent and ties it to the derivative being zero there. A stronger answer notices that the numerator factors as $(x - 3)(x + 1)$, so $x = -1$ is the only other point on the curve with a horizontal tangent. Do not accept “the function is zero there”: $f(3) = 9$, and it is the *slope* that vanishes, not the value.